

Table 25: Interpretation of Agents Decisions in Diverse Graph Learning Scenarios

Agents	Input information	Reason of decision	Agent output
Manager agent	User Request : I have a graph, it is saved on the path:Cor_a, in which node represent the paper and edges represent the citation relationship. The node attribute is the keywords mentioned in the paper.I want to predict the domain of the given paper.	{'Learning_tasks_on_graph_reason': 'The user wants to predict the domain of the given paper. The node attribute is the keywords mentioned in the paper. Therefore, the node-level learning task is chosen.', 'Learning_task_types_reason': 'The user wants to predict the domain of the given paper. Therefore, the classification learning task is chosen.', 'Evaluation_metric_reason': 'The user wants to predict the domain of the given paper. Therefore, the accuracy metric is chosen.'}	{'Data': 'Cor_a', 'Learning_tasks_on_graph': 'node-level', 'Learning_task_types': 'classification', 'Evaluation_metric': 'accuracy', 'Preference': 'None'}
Manager agent	User Request : In an environmental monitoring network at /data/environment_network/, nodes are sensor locations and edges represent ecological connections. The goal is to predict the overall health of an ecosystem. Graph-level attributes include biodiversity indices and pollution levels.	{'Learning_tasks_on_graph_reason': 'The goal is to predict the overall health of an ecosystem, which is a graph-level attribute. Therefore, a graph-level learning task is chosen.', 'Learning_task_types_reason': 'The goal is to predict a continuous variable (overall health), which is a regression task', 'Evaluation_metric_reason': 'R-squared is a common metric for regression tasks, and it is preferred because it provides a measure of how well the model fits the data.'}	{'Data': '/data/environment_network/', 'Learning_tasks_on_graph': 'graph-level', 'Learning_task_types': 'regression', 'Evaluation_metric': 'R-squared', 'Preference': 'None'}
Manager agent	User Request : In a book co-authorship network stored at /data/book_network/, nodes are authors and edges are co-authorships. The goal is to identify influential authors in different genres. I'm leaning towards using GCNConv.	{'Learning_tasks_on_graph_reason': 'The goal is to identify influential authors in different genres.', 'Learning_task_types_reason': 'The user wants to identify influential authors in different genres. Classification is used.', 'Evaluation_metric_reason': 'Accuracy is used as the evaluation metric because the user wants to identify influential authors in different genres.'}	{'Data': '/data/book_network/', 'Learning_tasks_on_graph': 'node-level', 'Learning_task_types': 'classification', 'Evaluation_metric': 'accuracy', 'Preference': 'GCNConv'}
Data agent	User Request : I have a graph, it is saved on the dataset file:Cor_a, in which node represent the paper and edges represent the citation relationship.The node attribute is the keywords mentioned in the paper.I want to predict the domain of the given paper. Task plan : {'Data': 'Cor_a', 'Learning_tasks_on_graph': 'node-level', 'Learning_task_types': 'classification', 'Evaluation_metric': 'accuracy', 'Preference': 'None'}	{'feature_engineering_reason': 'The AddSelfLoops transform is used to add self-loops to the given homogeneous or heterogeneous graph. This is useful for cases where the graph does not have self-loops, but the model requires them to be present.'}	{'feature_engineering': 'AddSelfLoops'}
Data agent	User Request : In an online gaming network at /data/gaming_network/, nodes are players and edges are interactions. The goal is to identify user engagement patterns. Node attributes include playtime and activity data. Task plan : {'Data': '/data/gaming_network/', 'Learning_tasks_on_graph': 'Node-level', 'Learning_task_types': 'Classification', 'Evaluation_metric': 'Accuracy', 'Preference': 'None'}	{'feature_engineering_reason': 'The gaming network is undirected, hence the ToUndirected feature engineering technique is suitable.'}	{'feature_engineering': 'ToUndirected'}
Configuration agent	Task plan : {'Data': 'Cor_a', 'Learning_tasks_on_graph': 'Node-level', 'Learning_task_types': 'Classification', 'Evaluation_metric': 'Accuracy', 'Preference': 'None'}	{'reason': 'The task is related to graph representation learning, and the hidden size cannot adopt the differentiable search algorithm. The design dimension does not contain any coarsening operations. Therefore, the differentiable search algorithm is suitable for this task.'}	{'algorithm': 'Differentiable Search'}